

## WASTEWISE SCAN · DESIGN DECISIONS REGISTER

# WasteWise Scan design decisions register

Design Molecule Lab · designmolecule.com

## DOCUMENT CONTROL

|             |                                             |         |                |
|-------------|---------------------------------------------|---------|----------------|
| Document ID | WSC-DEC-001                                 | Version | 0.3            |
| Status      | <b>Draft</b>                                | Date    | 2026-10-02     |
| Author      | Amish Chadha                                | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept               |         |                |
| Repository  | github.com/BoujeeEnjinia1701/wastewise-scan |         |                |
| Source      | docs/06-design-decisions.md @ uncommitted   |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                                                                                                                             |
|---------|------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.3     | 2026-10-02 | Amish Chadha | Decisions carried out: 2.0 mm walls in the model (R6 238 g, 258 g with the cap); paper band study result raised as an open decision on the extended InGaAs band                                                                                    |
| 0.2     | 2026-10-02 | Amish Chadha | Amish approved the recommendations for open items 1 to 7 on 2026-10-02 (WSC-DDR-004 accepted, R6 by thinner walls, SWaCH as first candidate partner, cooperative price, eight bands, QR pair ID, consented spectra terms); moved to decisions made |
| 0.1     | 2026-10-02 | Amish Chadha | Register opened with the build plan; budget treated as a value-engineering target                                                                                                                                                                  |

# WasteWise Scan design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan ( `docs/05-build-plan.md` ) describes the design as it stands and does not list open decisions.

## ■ Open decisions

| # | Decision                                                                                                                                                                                                                                                                                      | Options                                                                                                                                                                                                                                               | Recommendation                                                            | Source          |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------|
| 1 | Whether to add the extended InGaAs band now. The paper band study (WSC-CAL-001 v0.4, B4 to B7) gave 89.5 % correct with eight bands and 93 % for PE against PP at the assumed item scatter, above 98 % at 1.0 % scatter, so it neither shows nor rules out the problem that would trigger it. | (a) keep eight bands and settle it with the first labeled item set at TRL 4; (b) add a 1,700 to 1,750 nm band and an extended InGaAs photodiode now (extra cost not yet quoted; extended InGaAs photodiodes cost more than the USD 30 standard part). | (a): the result turns on item scatter, which only measured items can fix. | WSC-DDR-001, O3 |

## ■ To confirm when parts are bought

| # | What to confirm                                                                                                     | Why it matters                                                                                                                       | Source                    |
|---|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 1 | The protected cell's length (the cradle allows 69 mm) and the contact pair's thickness and spring travel            | The cradle end walls are 75 mm apart for a 69 mm cell, a 1 mm flat contact and a spring compressed to 4 mm plus its 1 mm plate       | WSC-DDR-004, P6           |
| 2 | The display board's real outline (about 62 x 26 x 9.5 mm is assumed), its connector positions and its battery input | The display frame and the 0.25 mm gap to the ceiling are set from the assumed outline; the slide switch is wired in the battery lead | WSC-DDR-004, P8           |
| 3 | Whether the board charges the cell with the switch in the battery lead, and from its 5 V pin                        | Charging needs the switch on; if the board cannot charge through its 5 V pin, the USB-C board wiring changes                         | Build plan, wiring        |
| 4 | The button's bezel (19 mm or less), its depth under the panel (14 mm or less) and its nut (20 mm or less across)    | They set the button hole and its clearance to the display frame and the amplifier board                                              | WSC-DDR-004, P9           |
| 5 | The slide switch's mounting hole pitch (15 mm) and body size (10 x 6 x 6 mm)                                        | The wall slot and holes are printed into the bottom shell                                                                            | WSC-DDR-004, P10          |
| 6 | Each LED's package: TO-46 (5.4 mm) or 5 mm lens LED (5.8 mm rim)                                                    | The head block pockets are 5.6 mm, for TO-46 cans                                                                                    | WSC-DDR-004, P4           |
| 7 | The foam gasket squeezes to about 0.6 mm with the bosses touching                                                   | The bosses stop the squeeze at 0.6 mm; a harder foam will not seal                                                                   | WSC-DDR-004, P1           |
| 8 | The photodiode's responsivity curve near 1,700 nm                                                                   | It decides how much of the 1,650 nm band is seen (R1)                                                                                | BOM line 4; WSC-CAL-001 B |

## ■ Value engineering

Value-engineering target: USD 150 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 167.50 (USD 17.50 over the target). Main cost drivers and savings worth trying:

- The largest lines are the eight LEDs with their drivers (USD 62; the six short-wave infrared LEDs cost about USD 8 to 15 each), the InGaAs photodiode (USD 30) and the display board (USD 20). Optics are 55 % of the cost.
- Making the design constructable added USD 3.50 net: the display frame, cell strap and gasket (lines 17 to 19) and more fixings; the bought cell holder became contacts in a printed cradle, which saved USD 1.
- Savings worth trying: the six-band option (dropping 1,050 and 1,300 nm saves about USD 18) if measured items show six bands suffice (WSC-DDR-001 D2; the paper band study scores six bands at 88 % against 89.5 % for eight, WSC-CAL-001 v0.4 B4); buying the LEDs as a set from one maker; a second source for the photodiode; at volume, a bare ESP32-S3 module and display in place of the display board.

■ Decisions made

| Date       | Decision                                                                                                                                                                                                                                                                                                                                                     | Decided by                                                                                                                                                                   | Record                                                 |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 2026-09-25 | D1 to D7: discrete LEDs with an InGaAs photodiode; USD 150 kept as the volume target with about USD 163 accepted for the prototype; T-Display-S3 class board; the eight-band set; a small on-device classifier with shared data; PTFE white reference in the cap; one protected 18650 cell                                                                   | Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."                                                    | WSC-DDR-001                                            |
| 2026-09-25 | D8 to D12: back clear items in sun with the black cap and refuse a result on through-light; black calibration cap; clip-on sun hood; open-air crosstalk reading in calibration; one shared scan record with a pair ID join                                                                                                                                   | Amish: "i accept all your recommendations, go with them across all repos."                                                                                                   | WSC-DDR-003,<br>WSC-DDR-002                            |
| 2026-09-26 | WasteWise Scan chosen as the pilot for product-grade renders; every drawing names the project and its repository                                                                                                                                                                                                                                             | Amish                                                                                                                                                                        | docs/<br>REVIEW.md ,<br>session<br>2026-09-26          |
| 2026-09-30 | Make the design physically buildable while drawing the build plan; outstanding decisions go in this register, not the build plan                                                                                                                                                                                                                             | Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." | WSC-DDR-004<br>(changes open for review, item 1 above) |
| 2026-10-01 | budget_usd is a value-engineering target, not a limit                                                                                                                                                                                                                                                                                                        | Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens."                                                             | This register,<br>Value engineering                    |
| 2026-10-02 | Design for construction accepted: the changes P1 to P12 of WSC-DDR-004, as made                                                                                                                                                                                                                                                                              | Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."                                    | WSC-DDR-004,<br>A2                                     |
| 2026-10-02 | Mass (R6): 2.0 mm shell walls, keeping the bosses and ribs, with the drop case checked at TRL 4; R6 stays not met until the first prototype is weighed (carried into the model 2026-10-02: 238 g, 258 g with the cap, 8 g over, since the shells saved 17 g rather than about 20 g), and the cap stays counted because it is carried for backing clear items | Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."                                    | WSC-DDR-004,<br>A1; WSC-CAL-001<br>[A3]                |
| 2026-10-02 | First partner and region: a member-owned waste picker cooperative that already sells sorted plastics to scrap buyers; the first candidate to approach is SWaCH in Pune, India, the partner chosen for WasteWise-ml, with Pune scrap prices as the first price table                                                                                          | Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."                                    | WSC-DDR-001,<br>O1                                     |
| 2026-10-02 | Value-engineering target: the USD 150 value-engineering target is the price a cooperative pays for a shared scanner, not an individual picker                                                                                                                                                                                                                | Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in                                                                                     | WSC-DDR-001,<br>O2                                     |

| Date       | Decision                                                                                                                                                                                                                                  | Decided by                                                                                                                                | Record                  |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
|            |                                                                                                                                                                                                                                           | after the review and aren't on the page either."                                                                                          |                         |
| 2026-10-02 | Extended InGaAs band: eight bands for the prototype; the paper band study on published reference spectra comes first, and the extended InGaAs band is added only if that study shows PE and PP cannot be told apart to R1 with eight      | Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either." | WSC-DDR-001, O3         |
| 2026-10-02 | Pair ID in the field: the scanner creates the pair ID and shows it as a QR code on its screen for the phone to scan, with the same short code in digits as a typed fallback                                                               | Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either." | WSC-DDR-002, question 2 |
| 2026-10-02 | Consented spectra: consented spectra join the WasteWise-ml field dataset on the same terms as field photos: opt-in per scan, co-owned by the pickers' organization, and published under CC BY 4.0 only with that organization's agreement | Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either." | WSC-DDR-002, question 3 |